

Standard FR4 TG135 Datasheet

Classification according to IPC-4101 E / 21

Reinforcement: Woven E-Glass
Resin System: Epoxy, unfilled



Explanations :

C = preconditioning in humidity chamber
E = preconditioning at temperature

The figures following the letter symbols indicate with the first digit the duration of the preconditioning in hours, with the second digit the preconditioning temperature in °C and with the third digit the relative humidity.

Laminate Requirements		Thickness < 0,50mm		Thickness ≥ 0,5mm		Units	Test Method
		Typical Value	Specification	Typical Value	Specification	Metric	IPC-TM-650 or as described
Peel Strength, minimum	A: Low profile copper foil and very low profile copper foil – all copper foil > 17µm	0,9	0,70	0,95	0,70	N/mm	2.4.8
	B: Standard profile copper foil						
	1. After thermal stress	1,05	0,80	1,20	1,05		2.4.8.2
	2. At 125°C	0,95	0,70	1,15	0,70		2.4.8.3
	3. After process solutions	0,8	0,55	1,0	0,80	AABUS	2.4.8
	C: All other foil - composite		AABUS		AABUS		
Volume Resistivity, minimum	A: C-96/35/90	4 10 ⁵	10 ⁵	6 10 ⁵		MΩ cm	2.5.17.1
	B: After humidity conditioning			4 10 ⁵	10 ⁶		
	C: At elevated temperature	7 10 ⁵	10 ³	7 10 ⁵	10 ³		
	E-24/125						
Surface Resistivity, minimum	A: C-96/35/90	1 10 ⁵	10 ⁴			MΩ	2.5.17.1
	B: After humidity conditioning			3 10 ⁵	10 ⁴		
	C: At elevated temperature	6 10 ⁵	10 ³	6 10 ⁵	10 ³		
	E-24/125						
Moisture Absorption, maximum		0,4		0,4	0,80	%	
Dielectric Breakdown, minimum				45	40	kV	2.5.6